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ICOM5015 001D, Artificial Intelligence

Date: April 16, 2026

Is Artificial Intelligence True Intelligence?

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1. Problem Statement

The central question of this assignment is whether artificial intelligence (AI) constitutes genuine intelligence comparable to human intelligence. Addressing this question requires first clarifying what is meant by “intelligence,” since the answer depends heavily on how the concept is defined and interpreted across different disciplines.

Intelligence is commonly understood as the capacity to learn from experience, reason information, solve problems, and apply knowledge in adaptive and meaningful ways. In humans, however, intelligence extends beyond these functional abilities. It includes consciousness, self-awareness, intentionality, and subjective experience. These elements are studied in fields such as cognitive science, philosophy of mind, and neuroscience, and they suggest that human intelligence is not only behavioral but also experiential and internally grounded.

Artificial intelligence, by contrast, refers to computational systems designed to perform tasks that typically require human intelligence. These tasks include language processing, pattern recognition, decision-making, and problem-solving. Modern AI systems-particularly those based on machine learning-operate by analyzing large amounts of data and identifying statistical patterns that allow them to produce appropriate outputs. For example, large language models can generate human-like text by predicting the most likely sequence of words given prior context, rather than by understanding meaning in the way humans do.

Within the framework of AI, systems are often modeled as rational agents that act to achieve specific goals by optimizing performance measures. From this perspective, intelligence can be seen as the ability to produce successful outcomes in each environment. However, this raises a deeper philosophical issue: is intelligence merely the ability to behave intelligently, or does it require genuine understanding, awareness, and intentional mental states?

The central issue, therefore, is whether performing intelligent behavior is sufficient to qualify as true intelligence, or whether deeper properties, such as understanding, consciousness, and intentionality, are necessary conditions. This distinction lies at the heart of ongoing debates in philosophy and artificial intelligence, and it motivates the examination of arguments both for and against the claim that AI constitutes real intelligence.

2. Review of Arguments

Position A: AI is True Intelligence

One of the primary arguments supporting the view that artificial intelligence constitutes real intelligence is grounded in functionalism. According to this perspective, mental states are defined by what they do rather than by the material from which they are made. If a system can reason, learn, and solve problems, then it qualifies as intelligent regardless of whether it is biological or artificial. As explained in [1], cognition depends on functional organization, not physical composition. This implies that artificial systems, if functionally equivalent to human cognition, should be considered genuinely intelligent.

A second argument is that human intelligence itself develops through processes similar to those used in artificial intelligence systems. Psychological research shows that humans learn through observation, imitation, and reinforcement. Albert Bandura's Social Learning Theory explains that learning emerges from interaction with the environment rather than from immediate understanding [2]. Humans, especially in early development, rely heavily on mimicry and feedback before achieving deeper comprehension. This mirrors how AI systems are trained: they analyze data, identify patterns, and improve performance through feedback mechanisms. Therefore, if human intelligence can emerge from pattern-based learning, then AI systems following similar processes can also be considered genuinely intelligent.

Another argument challenges the idea that intelligence must be tied exclusively to biological systems. Scientific research suggests that intelligence can arise from different physical substrates. For example, researchers demonstrated that biological neurons grown in a lab could learn to play a simple video game, illustrating that intelligence can emerge from non-traditional systems [3]. This supports the concept that intelligence is not dependent on a specific material, such as the human brain, but rather on the organization and interaction of components. If intelligence can exist in both biological and artificial systems, then restricting it to humans becomes arbitrary.

Finally, proponents highlight the phenomenon known as the AI effect. Historically, tasks such as playing chess, solving mathematical problems, or translating languages were once considered clear indicators of intelligence. However, once AI systems achieved these capabilities, they were often reclassified as non-intelligent processes. This shifting definition suggests that the boundary of intelligence is not fixed but continuously adjusted as technology advances [4]. Rather than proving that AI lacks intelligence, this pattern indicates that human standards of intelligence evolve to exclude artificial systems.

Taken together, these arguments support the conclusion that intelligence is best understood as a functional and emergent property. If artificial systems can learn, adapt, and perform complex cognitive tasks through mechanisms comparable to human processes, then AI can reasonably be considered a genuine form of intelligence.

Position B: AI is Not True Intelligence

From Position B, it is argued that artificial intelligence does not constitute genuine intelligence because it lacks key characteristics that define human cognition, particularly understanding, consciousness, and intentionality. While AI systems can replicate intelligent behavior, these arguments maintain that such behavior does not imply the presence of real mental states.

First, AI does not truly understand the meaning. This argument is most famously articulated by philosopher John Searle through his Chinese Room thought experiment. In this scenario, a person who does not understand Chinese is placed inside a room and given a set of rules for manipulating Chinese symbols. By following these rules, the person can produce responses that appear meaningful to an outside observer, even though they do not understand the language. Searle

uses this example to argue that computational systems operate purely on syntax, the manipulation of symbols based on formal rules, without access to semantics or meaning. Similarly, AI systems process inputs and generate outputs based on statistical patterns and programmed algorithms, but they do not genuinely comprehend the content they are handling. Even advanced systems that produce fluent and coherent language do so without any awareness or understanding of what the words mean.

Second, AI lacks consciousness. As explained by philosopher David Chalmers, there is a fundamental distinction between explaining behavior and explaining subjective experience. While science and engineering can account for how systems process information and produce outputs, they do not explain why or how conscious experience arises. This is often referred to as the “hard problem” of consciousness. AI systems, regardless of their complexity, do not possess subjective awareness or inner experiences. They do not feel, perceive, or experience anything; they simply execute computational processes. As a result, even if an AI system behaves in ways that resemble human intelligence, it lacks the conscious dimension that is central to human cognition.

Third, AI lacks genuine intentionality. Intentionality refers to the capacity of mental states to be about something, to have beliefs, desires, or goals directed toward objects or states of the world. Philosopher Daniel Dennett suggests that we can adopt an “intentional stance” toward systems, treating them as if they have beliefs or desires to predict their behavior. However, this is a useful interpretive strategy rather than evidence of real mental states. AI systems do not form goals independently or possess intrinsic motivations; they operate according to objectives defined by their designers and training processes. Their “goals” are externally imposed rather than internally generated, which distinguishes them from genuinely intentional agents.

These arguments collectively support the conclusion that AI systems simulate intelligence rather than possess it. While they may perform tasks associated with intelligent behavior, sometimes even surpassing human performance, they do so without understanding, consciousness, or true intentionality. Therefore, according to Position B, artificial intelligence should be understood as a powerful tool that mimics aspects of human intelligence, rather than an entity that genuinely possesses it.

3. Discussion & Analysis

Strengths of Position A

A key strength of Position A is that it relies on a clear and operational definition of intelligence. By focusing on function, such as learning, reasoning, and problem-solving, it allows intelligence to be evaluated in observable and measurable terms rather than abstract qualities [1]. This makes the argument applicable across different systems, including AI.

Another strength is the parallel between human and machine learning. The idea that humans develop intelligence through imitation and reinforcement, as described in the Social Learning Theory [2], provides a strong conceptual bridge between biological and artificial systems. This comparison makes AI less of an outlier and more of a continuation of known learning processes.

Position A is also strengthened by its consistency with scientific perspectives. Evidence that learning can emerge from different physical systems supports the idea that intelligence is not tied exclusively to human biology [3]. This reinforces the claim that intelligence can exist in multiple forms.

Finally, the AI effect highlights an important methodological issue: the tendency to redefine intelligence as machines achieve it [4]. This strengthens Position A by questioning whether criticisms of AI are based on stable criteria.

Weaknesses of Position A

One limitation of Position A is that it may rely heavily on observable behavior as the main indicator of intelligence. While this provides clarity, it may not fully address whether internal processes, such as understanding or awareness, should also be considered part of intelligence.

Additionally, the comparison between human and AI learning, while useful, may abstract away important differences. Human learning often develops into rich conceptual understanding, and it is not fully clear whether current AI systems reach that same level.

The argument that intelligence is independent of physical substrate is compelling, but it may leave open questions about whether all implementations of intelligence are equivalent, or whether some forms include additional properties not yet present in AI.

Finally, the AI effect argument mainly challenges how intelligence is defined, but it does not by itself establish what criteria should replace those shifting definitions, leaving some ambiguity in how intelligence ought to be evaluated.

Strengths of Position B

A major strength of Position B is its emphasis on understanding as a core component of intelligence. By distinguishing between producing correct outputs and genuinely understanding them, it introduces a deeper level of analysis beyond performance.

The distinction between syntax and semantics further strengthens this position. It provides a clear framework for explaining why systems that manipulate symbols may still lack meaning, highlighting a potential limitation of current AI systems.

Position B also benefits from focusing on qualitative aspects of intelligence, such as consciousness or subjective experience. These aspects are widely associated with human cognition and raise important questions about whether intelligence should include more than functional ability.

Weaknesses of Position B

One potential limitation of Position B is that it depends on a more restrictive definition of intelligence, which may exclude systems that demonstrate many traditionally intelligent behaviors.

The Chinese Room argument raises important concerns about understanding, but it assumes that intelligence must involve human-like comprehension, which may not account for alternative forms of intelligence.

Additionally, while the position highlights what AI may lack, it provides less clarity on how to evaluate intelligence in systems that clearly learn and adapt, leaving some tension between theory and observable evidence.

Finally, responses such as the systems' reply suggest that understanding might emerge at a broader level, indicating that the issue is still open to interpretation rather than conclusively resolved.

4. Reflection

The debate had a significant impact on our understanding of the question of whether artificial intelligence constitutes real intelligence. In the interest of transparency, we admit every member of the group was initially more aligned with Position B, the view that AI is not true intelligence. This initial stance was largely influenced by the intuitive belief that intelligence requires understanding, consciousness, or human-like awareness, qualities that AI systems do not clearly possess.

Through the process of research, discussion, and engaging with opposing arguments, our perspectives began to understand Position A. Examining theories such as functionalism and comparing human learning processes with machine learning challenged our initial assumptions about what intelligence fundamentally is. In particular, the realization that human intelligence itself develops through mechanisms like imitation, pattern recognition, and feedback made the distinction between human and artificial intelligence less clear-cut than we had originally assumed.

The debate format, especially the requirement to fairly represent and defend both positions, played a crucial role in this shift. By engaging seriously with Position A, we were forced to move beyond intuitive reactions and instead evaluate the issue using consistent criteria and evidence. This process revealed that some of our earlier views relied on implicit assumptions about intelligence that were not as well-defined or universally accepted as we had thought.

At the same time, the arguments from Position B remained compelling, particularly those emphasizing understanding and meaning. Rather than completely abandoning our initial perspective, the debate led us to adopt a more nuanced view that recognizes the strengths and limitations of both sides.

5. Conclusion

The question of whether artificial intelligence constitutes real intelligence ultimately depends on how intelligence itself is defined. Throughout this essay, two competing perspectives were examined. Position A argues that intelligence should be understood in functional terms, emphasizing the ability to learn, adapt, and solve problems regardless of the underlying material. From this view, artificial intelligence qualifies as a genuine form of intelligence because it demonstrates many of the same capabilities observed in human cognition. In contrast, Position B

maintains that true intelligence requires deeper elements such as understanding, meaning, or consciousness, which AI systems do not clearly possess.

The analysis of both positions reveals that each provides valuable insights but also leaves important questions unresolved. Functional approaches offer clarity and consistency but may overlook internal aspects of cognition, while arguments centered on understanding highlight meaningful distinctions but rely on concepts that are difficult to define and measure. As a result, neither perspective fully resolves the issue, but together they expose the complexity of defining intelligence itself.

Considering this analysis, the group ultimately aligns more closely with Position B. While artificial intelligence demonstrates increasingly sophisticated capabilities, it does not yet exhibit the level of understanding or awareness required to be considered fully equivalent to human intelligence. This conclusion reflects the current state of technology rather than a definitive limitation of artificial systems.

At the same time, this is not a question with a fixed or final answer. As artificial intelligence continues to advance, the criteria we use to define intelligence may continue to evolve. This makes the boundary between “real” intelligence and artificial systems increasingly difficult to define. For this reason, we conclude that this debate is inherently ongoing and will likely become even more complex as AI systems continue to develop and challenge our existing definitions of intelligence.

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